DataDive Week 4:
ML Models
Understanding The Basics

SheffDataSoc

Data Science Society

Datasoc Promo!

Upcoming social: **Pub Quiz @ the Cavendish** (West St), **Thu 12th March 7-10pm**. It's free to attend, so please do come along!

Check out Datasoc on instagram (**@sheffdatasoc**) for updates and information on our events!

Tell your friends and classmates about the Data Dive! It's a fun way to work on a project, get experience for your CV, and win actual prizes!

Session Outline

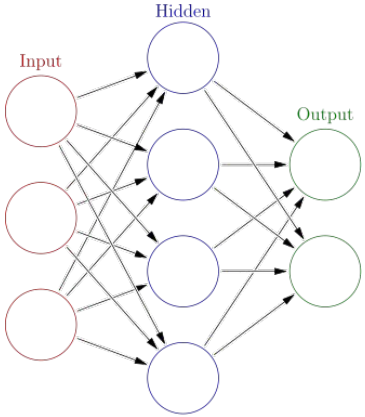
- What is an AI model?
- What kind of problems can we solve?
- What types of models are there?
- Why do they work?
- How do we train them?

Access the resources for today's session:

https://github.com/sheffdatasoc/datadive-resources

What can AI do?

AI models solve problems by **learning and exploiting patterns in data**



Dog

An AI model is only as good as the patterns it has to exploit. If there is no pattern, we shouldn't use AI! This is important to think about when deciding if your problem is solvable with AI.

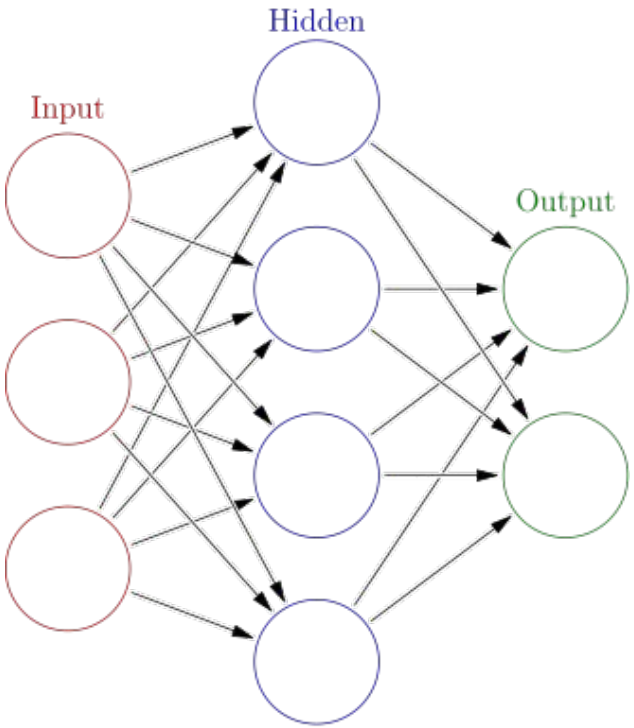
What is an AI model?

Models are mathematical systems that predict an output based on an input.

Every ML model has three things:

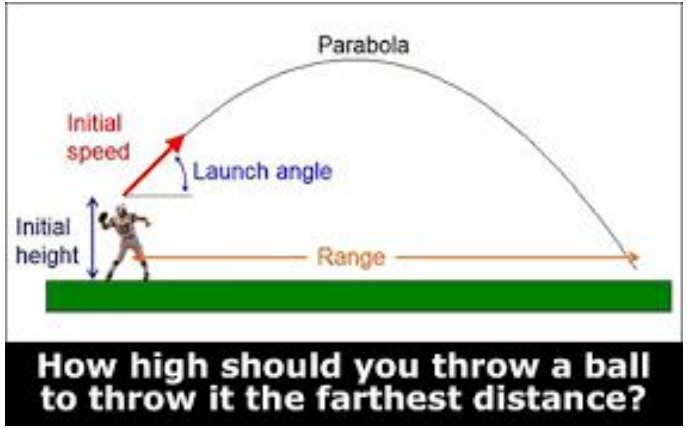
1. Weights
2. Loss function
3. A way to find good weights

Information flows through the network, transformed at each step, eventually producing an output. The input is transformed by the weights - this is why we must find weights that are good for our goal. The loss function is an ‘objective’ that helps us find good weights.



The Core Ability

The key intuition for understanding models is that they are **function approximators**



Nearly everything can be modelled as a function!

Why Models Work

We have a set of data and we want to predict something with it. We want to **approximate the function** that does this.

A simple model will struggle to approximate a complex function, but as we make it larger we give the model **more power to approximate well**.

Model Size ≈ Approximation Strength

In fact, **if we have a big enough model we can model any function**!

Deep Learning vs Machine Learning

You may have heard of these terms. What do they actually mean?

In Machine Learning, we typically give the model specific features to look for, such as edges, colours and shapes. In Deep Learning, we just give the model the input and let it work out all these features itself.

Well, why don’t we always use Deep Learning then?

Deep Learning vs Machine Learning

AI models are split into two categories, but which should you choose?

	Deep Learning	**Machine Learning**
Problem Solving Domains	Language, Images, Audio, Video	Structured Data (e.g. housing prices, stock market)
Training	Slow, Expensive	Quick, Cheap
Inference	Slow, Expensive	Quick, Cheap
Generality	Often Very High	Usually Specialised

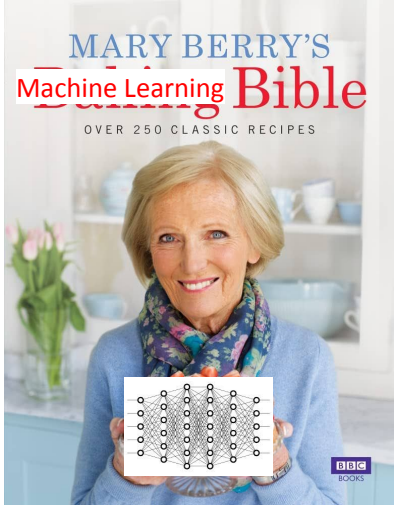
There is no correct answer. **Choose the model that best serves your problem**

A Data Scientist's Cookbook

All ML and DL models contain these ingredients:

1. **Data**
2. **Model Architecture**
3. **Objective (a way to find best approximation)**

Any bad selection of one of these ingredients can kill the model's performance. We must be careful.



The Objective: Loss Function

We need to define an objective for the model - something to tell it how to adjust itself so it performs in the way we want it to.

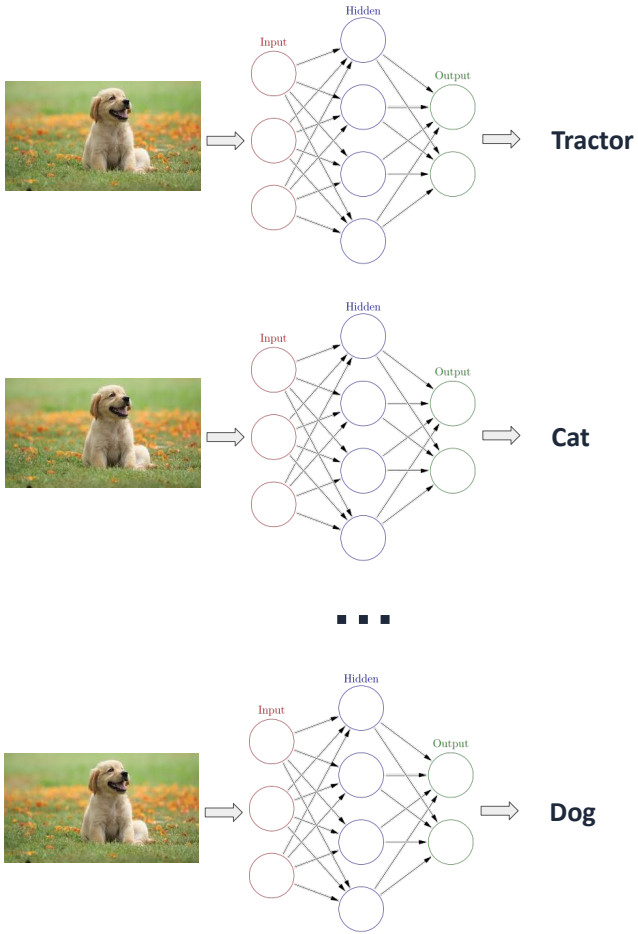
For example, if we want to classify dogs vs cats, we will reward it when it is correct and punish it when it fails. In practice, these rewards and punishments are just updates to the weights of the model.

The main tool for doing this is a **loss function**. This tells the model how wrong it is.

Training Overview

1. Start with random weights
2. Make predictions → compare to actual answers → measure error
3. Adjust weights to minimise error
4. Repeat until the network gets better

Goal: minimise prediction error on training data

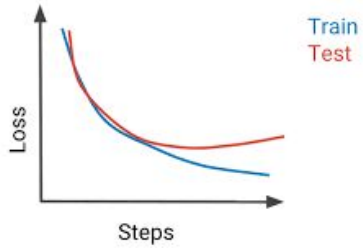


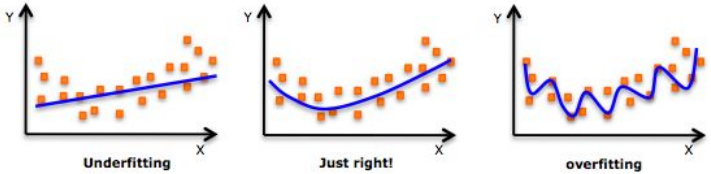
Practical Tips for Training

Thankfully, the infrastructure for training models is very mature. We will be using the library ‘TensorFlow’, from Google, a toolkit that gives us everything we need to train a model.

Some practical tips:

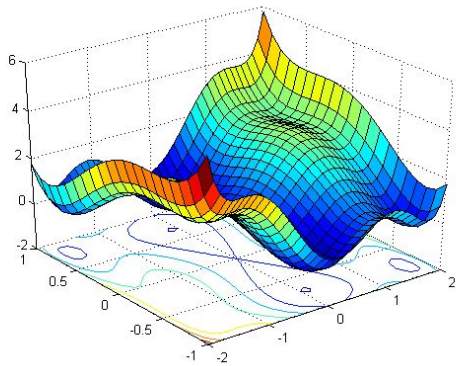
1. We must split our data into ‘test’ and ‘train’ sets
2. We must be careful to not overfit or underfit
3. We must evaluate the training progress at regular intervals





The Mechanics of Training

Recall that the goal of training a model is to minimise the loss.



Each point on the object is a combination of weights. The lower down on the graph, the lower the loss. Therefore, we want to find the bottom.

If we find the bottom, we have found the best set of weights and therefore the best approximation of our function! The model will be strong.

The Mechanics of Training

How do we find the bottom? Gradient Descent.

Imagine you're blindfolded on top of a mountain and need to get the bottom.

You:

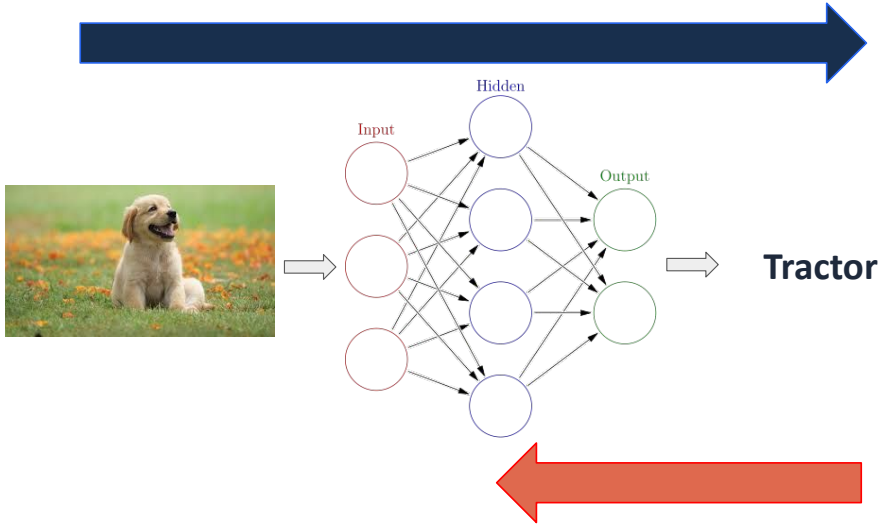
1. Find the steepest direction from where you're currently stood
2. Take a step that way
3. Repeat until flat (at the bottom)

This is exactly how neural networks learn!

The Training Cycle

Forward Pass

1. Put some data into the network
2. Make a prediction
3. Compare to the truth



Backward Pass

1. Find the steepest downward direction of the loss surface
2. Take a step
3. Update the weights
4. Reduce the error

Key Takeaways

Models

Different models are good at different things

Nearly everything is a function

Models can approximate any function

Data + Architecture + Objective = Good Model

Training

The forward-pass backward-pass cycle is how models learn

Gradient descent finds the best set of weights

A model is only as good as its ingredients

Where Do We Go From Here?

Model Evaluation - How do we know if our model is actually good?

Real-world applications & limitations

Ethical Evaluation of models

What comes next?

Thank you for coming! Make sure to spread the word and get your friends involved in the Data Dive!

You can still sign up if you are considering becoming a project lead:

https://tinyurl.com/datadiveprojectlead

Week	Session	Where/When
5	Evaluating & Interpreting ML Results and Team Formation & Sign-Up!	**Wednesday, 4-6pm Diamond LT9**

You will have the chance next week to group and form teams if you haven't already, so come along!